You are trying to control the rotational speed of a motor by using a PI controller. The goal of the controller is to maintain 10 rotation/s. You have measured the resulting gain and oscillation period of the system 0.6 and 10 respectively.

[error, e = required value - actual value]

Here the vertical axis represents the speed value of the motor and the horizontal axis represents the steps.

a. Calculate value of the Kp and Ki parameters of the given controller.

Answer:

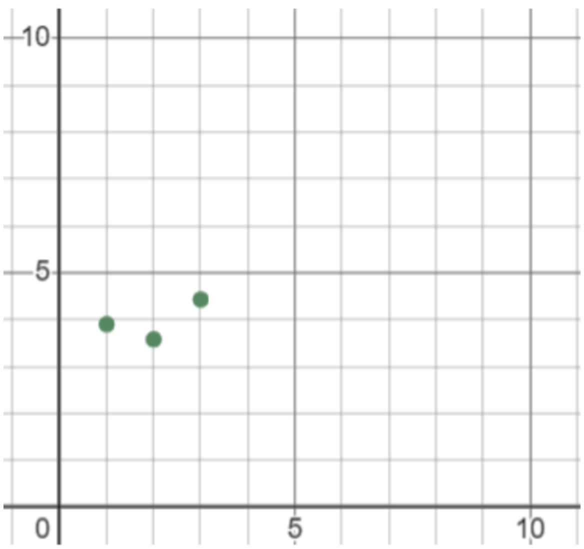
for PI control

$K_p = 0.45\,K$

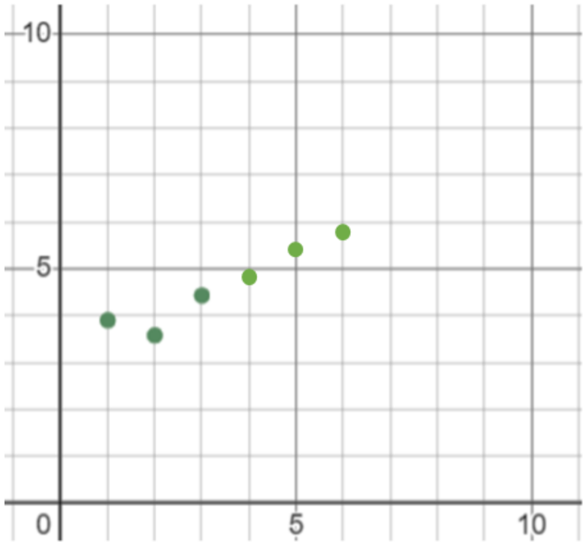
$K_i = 1.2 / P$

Kp = 0.45 * 0.6 = 0.27
Ki = 1.2 / 10 = 0.12

b. In the given diagram, speed values of the first three steps are given in the vertical axis by using the calculated parameters, and simulate the next three speed values of the motor step by step. You can plot on the question paper but the calculations should be shown in your answer script.



Answer:
Rotation Speed for t1= 3.9
Rotation Speed for t2= 3.6
Rotation Speed for t3= 4.3
$Error_1$= 10 – 0 = 10
$Error_2$= 10 – 3.9 = 6.1
$Error_3$= 10 – 3.6 = 6.4
$Error_4$= 10 – 4.3 = 5.7
Output = Kp * error (required value - actual value) + Ki * (error1 + error2 + … +errorn (0))
Rotation Speed for t4= 0.27 * 5.7 + 0.12*(10 + 6.1 + 6.4 + 5.7) = 4.923
$Error_5$= 10 - 4.923 = 5.077
Rotation Speed for t5 = 0.27 * 5.077 + 0.12*(10 + 6.1 + 6.4 + 5.7 + 5.077) = 5.36403
$Error_6$= 10-5.36403= 4.63597
Rotation Speed for t6= 0.27 * 4.63597 + 0.12*(10 + 6.1 + 6.4 + 5.7 + 5.077 + 4.63597) = 5.8012683


10
5
0
5
10